



Department of Computer Science

Assignment#4

Course Code: CS-121

Course Title: Digital Logic Design

Total Marks: 15

Assigned Date: 13th June, 2025.

Name: _____

Registration No. _____

Discipline/Semester: BS(CS)- 2nd B

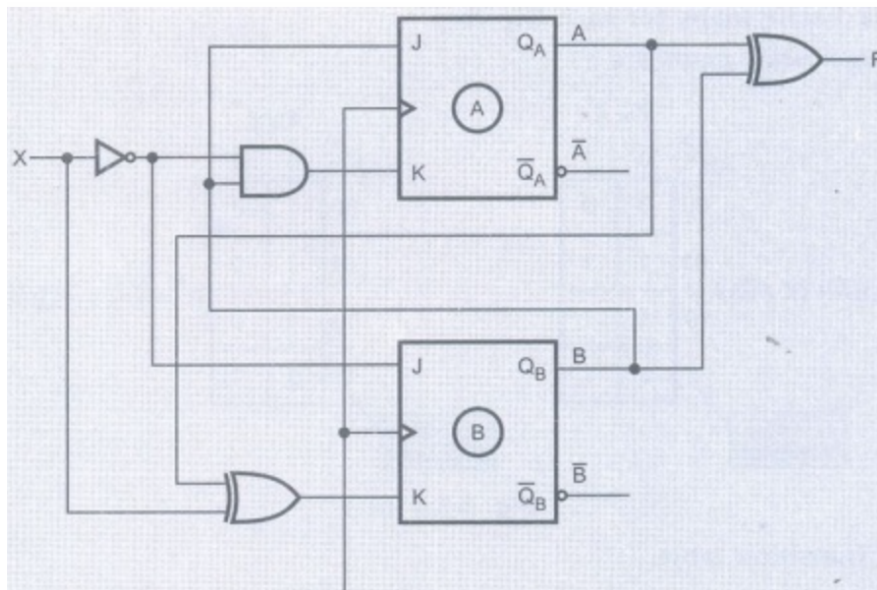
Submission Date/Deadline: 18th June, 2025.

Instructions:

- Late submissions will not be accepted.
- Hand written assignments will be accepted only

CLO3-GA4-BT-Level C4

Q1. Construct the transition table, state table and state diagram for the sequential circuit.



Q2. A sequential circuit with 2D Flip Fs A and B and input X and output Y is specified by the following next state and output equations.

$$A(t+1) = AX + BX$$

$$B(t+1) = A'X$$

$$Y = (A+B) X'$$

- Draw the logic diagram of the circuit.
- Derive the state table.
- Derive the state diagram.

